6

ME1100 Thermodynamics

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Gibbs Phase Rule

F=c-φ +2

F: degrees of freedom
c: number of components
φ: number of phases

Water liquid and water vapour(steam) are in **thermodynamic equilibrium**

F=1-2+2=1

Thus, we will see that specifying temperature will fix pressure or vice versa.
e.g. we know that water boils at 100 °C under 1 atmosphere of pressure.

If, ice, water liquid and vapour are in **thermodynamic equilibrium**

F=1-3+2=0

Vapour liquid equilibrium of pure substance

- A certain amount of liquid is contained in a container maintained at pressure P and temperature T. Molecules in the liquid possess kinetic energy and are in constant motion. Those molecules near to the liquid surface can "escape" (evaporate) into the vapour phase above the liquid.

- As time passes, there will be more and more molecules from the liquid phase that enters the vapour phase. Some of these molecules in the vapour phase will collide with one another, and some will "re-enter" (condense) into the liquid phase.

- Eventually the vapour space above the liquid will become saturated with molecules, that the tendency of molecules to escape (vapourise) is just balanced by the tendency for them to be recaptured (condensed). At this point, we say that an equilibrium condition had been reached between the vapour phase and the liquid phase at the given pressure P and temperature T.

http://www.separationprocesses.com/Distillation/DT_Chp01k.htm

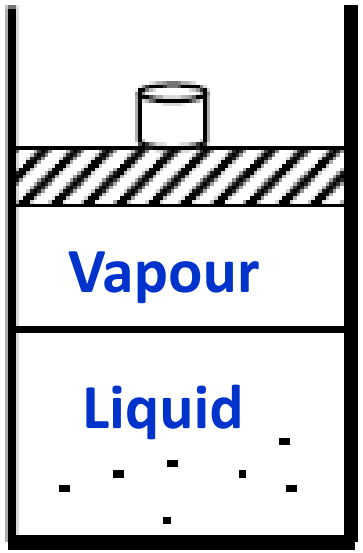
- The pressure exerted by the molecules in the vapour phase on the walls of the container is known as the equilibrium vapour pressure (PVP) of the liquid at temperature T. The equilibrium vapour pressure of the liquid is a strong function of temperature, and as the temperature increases the equilibrium vapour pressure increases.

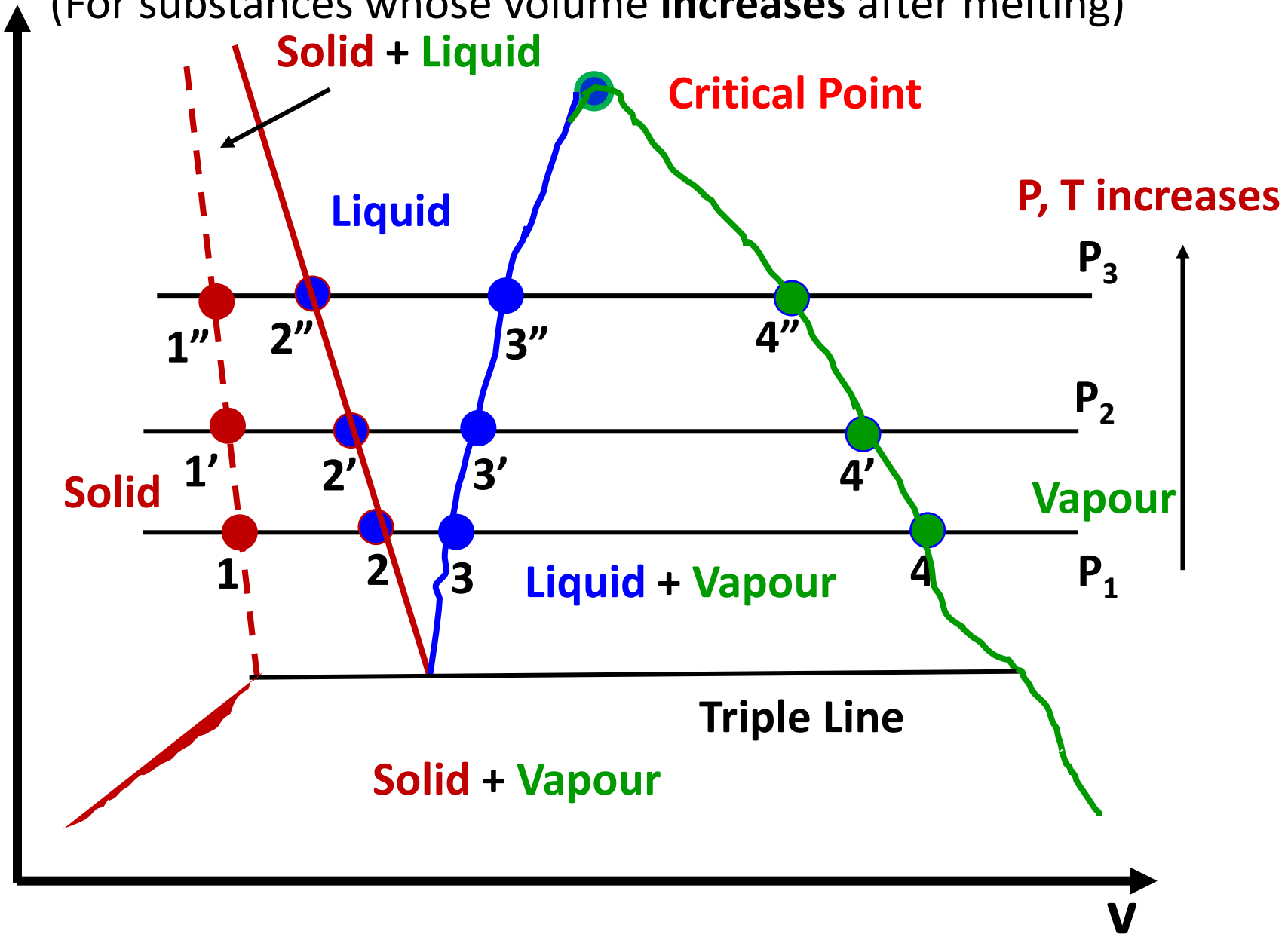
- The temperature at which the equilibrium vapour pressure exerted by the liquid equals the atmospheric pressure is the normal boiling point of the liquid.

- When the pressure is increased, it will be more difficult for the liquid molecules to escape into the vapour phase. Thus, the boiling point of a liquid increases with pressure; and the reverse is true for lower pressure.

Change of Phase: heating at a constant pressure

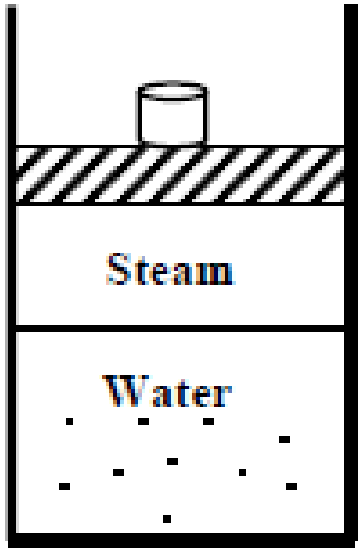
(For substances whose volume **increases** after melting)

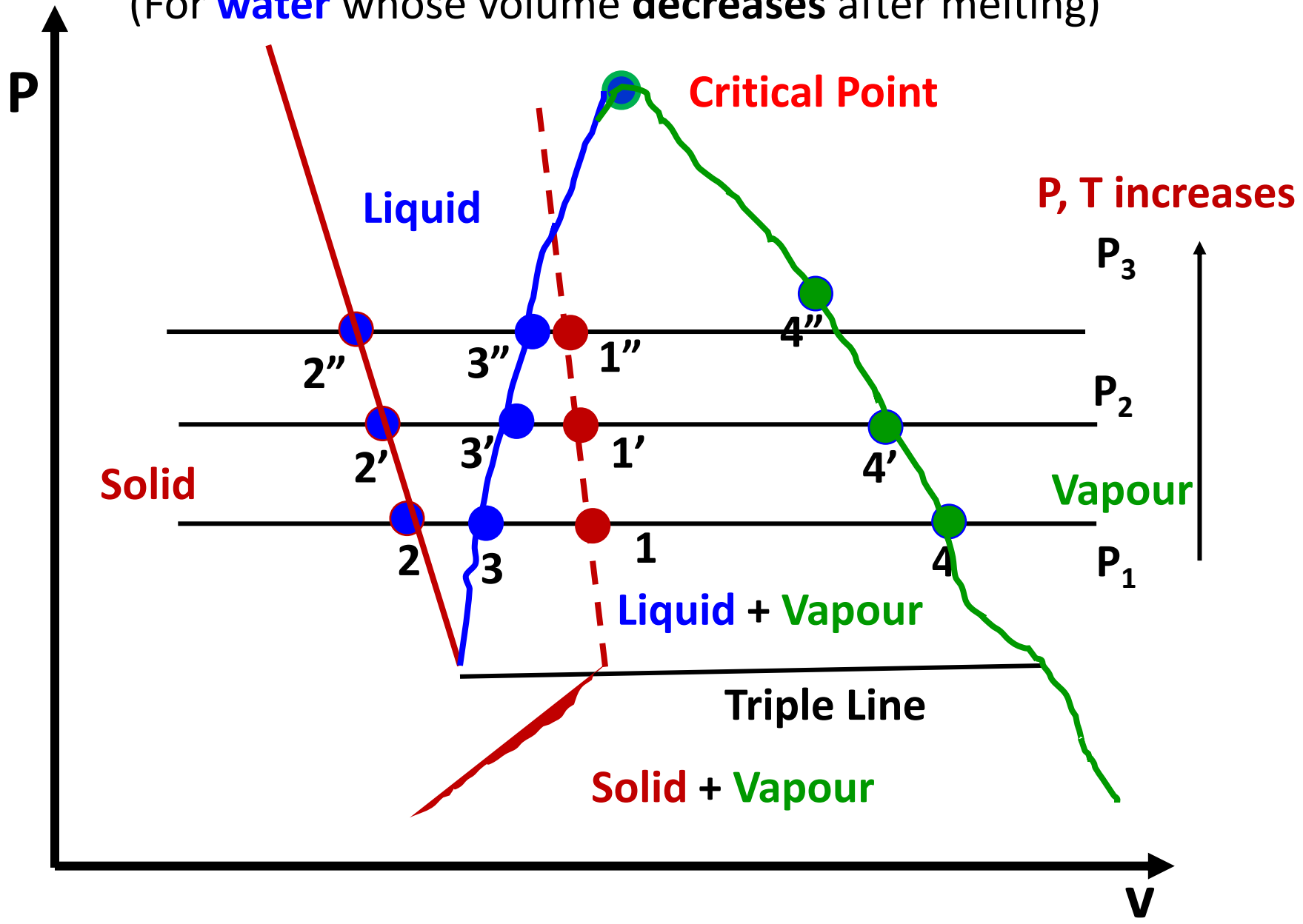




Change of Phase: heating at a constant pressure

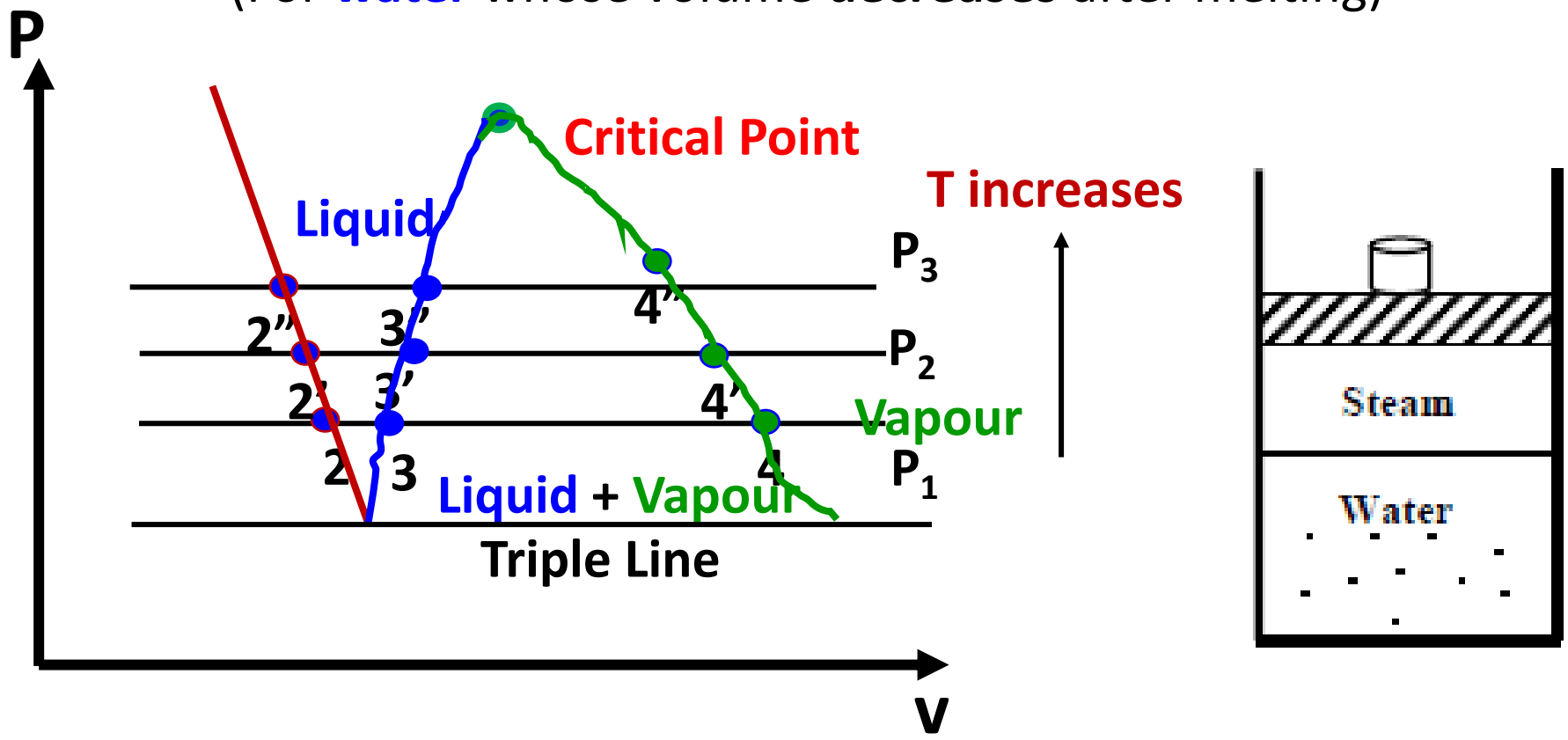
(For **water** whose volume **decreases** after melting)





Change of Phase: heating at a constant pressure

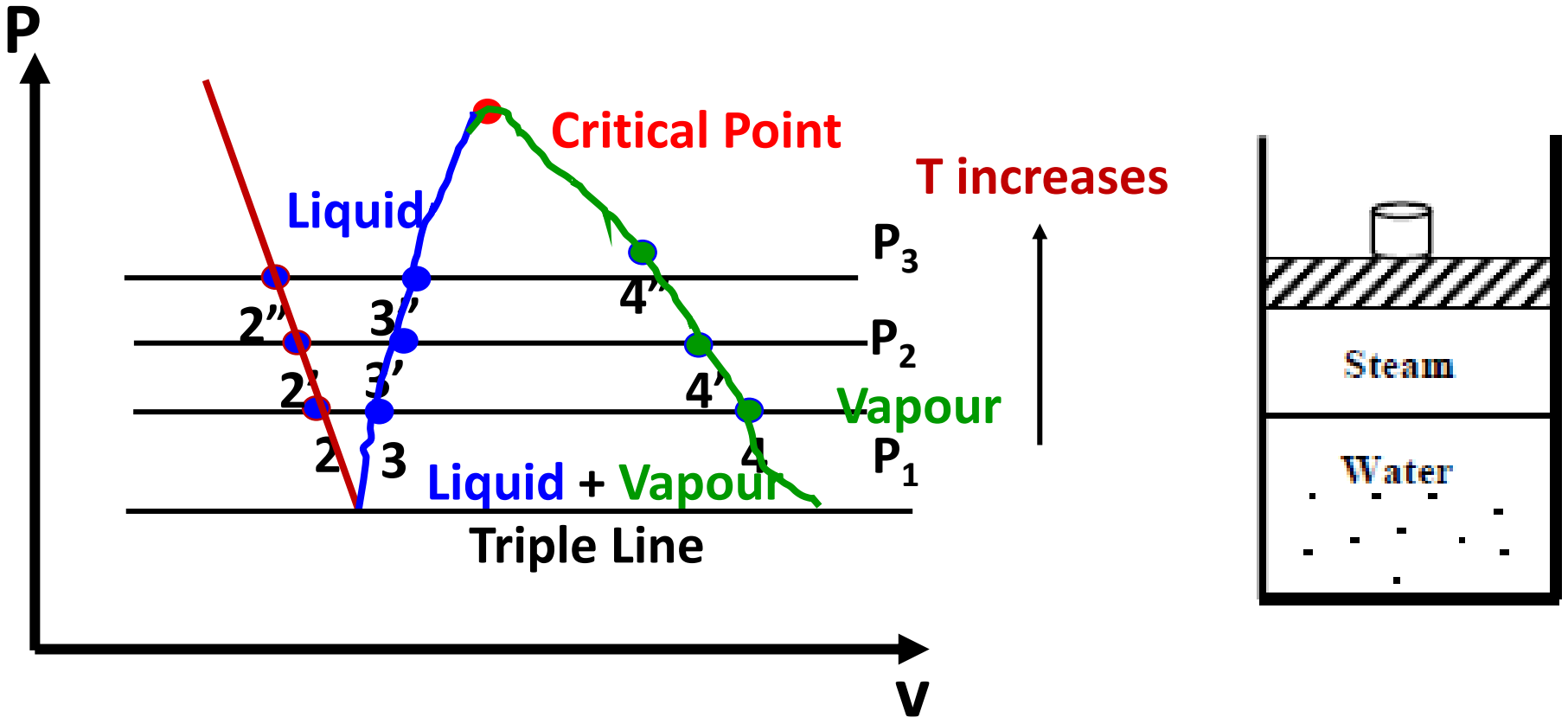
(For **water** whose volume **decreases** after melting)



A critical point, also called a critical state, specifies the conditions (temperature, pressure) above which distinct liquid and vapour phases cease to exist.

Water: **647 K (374 °C) and 22.064 MPa.**

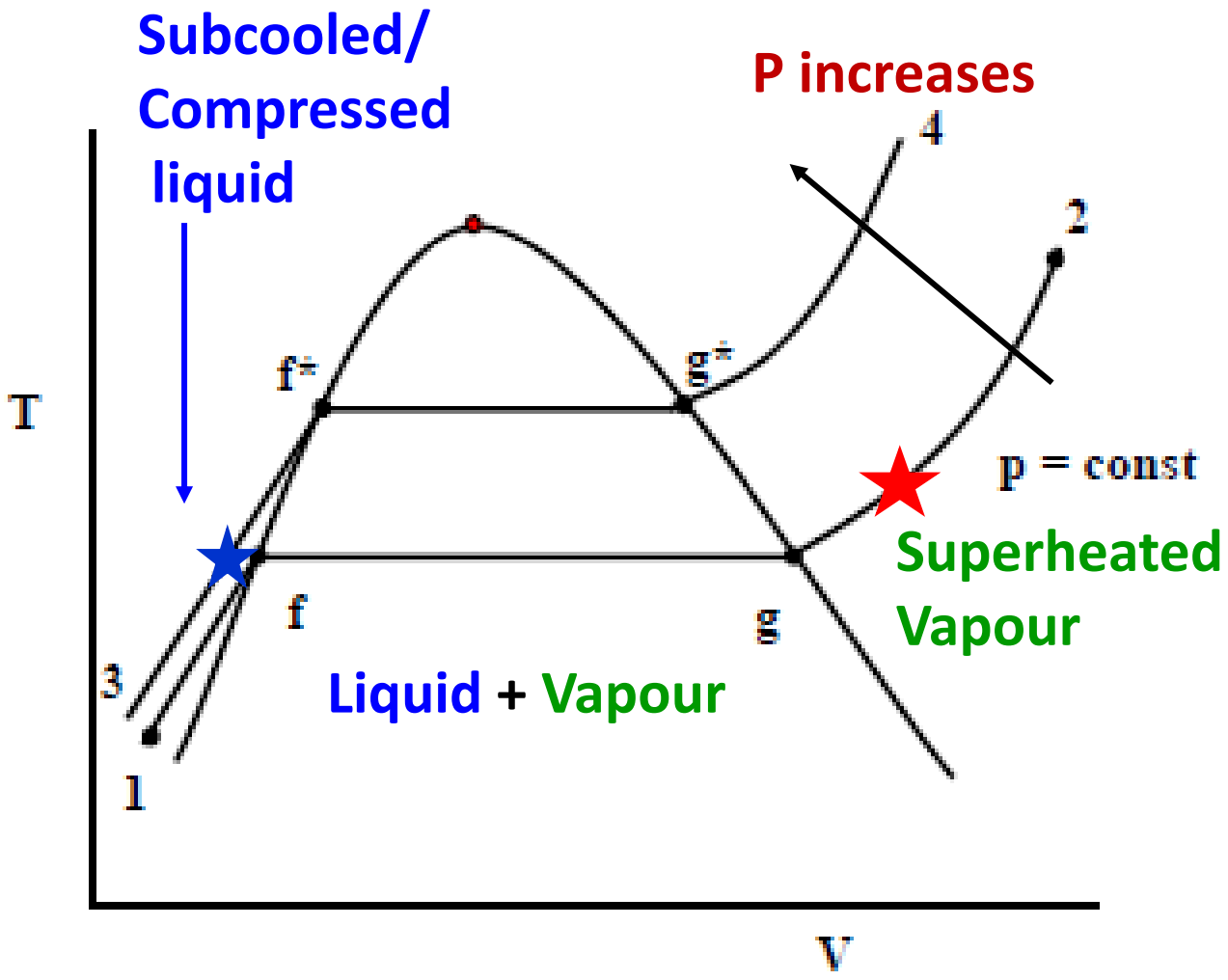
Change of Phase : heating at a constant pressure

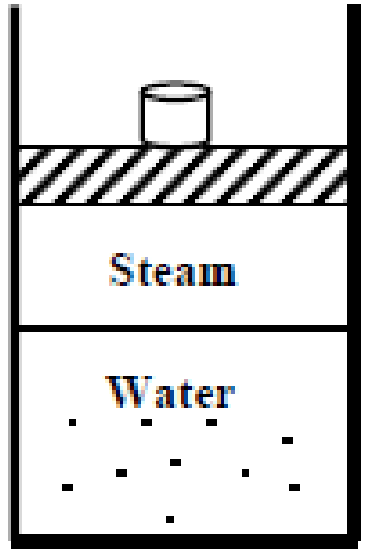


Triple point of a substance is the temperature and pressure at which three phases (gas, liquid, and solid) of that substance may coexist in thermodynamic equilibrium.

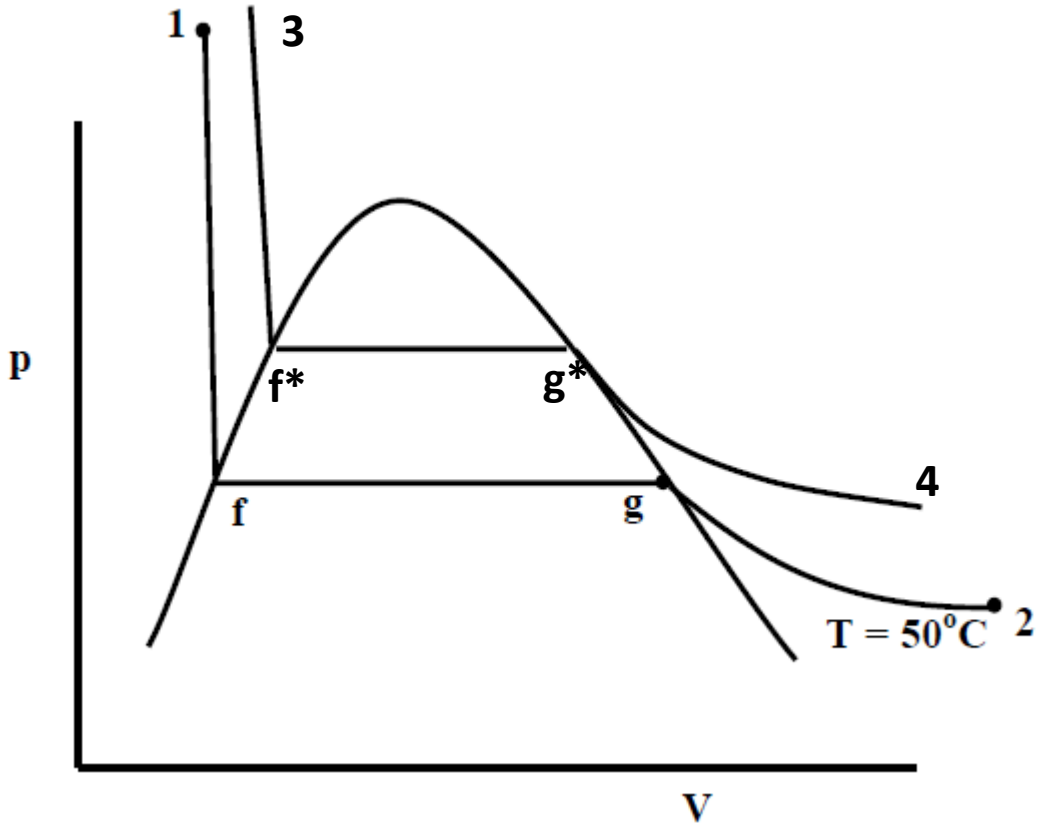
Water: 273.16 K, 611.66 Pa

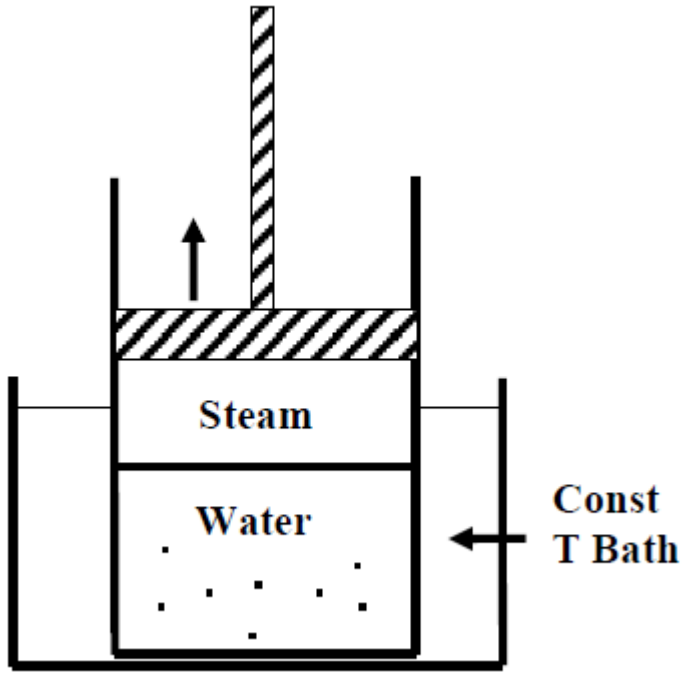
Change of Phase: heating at a constant pressure

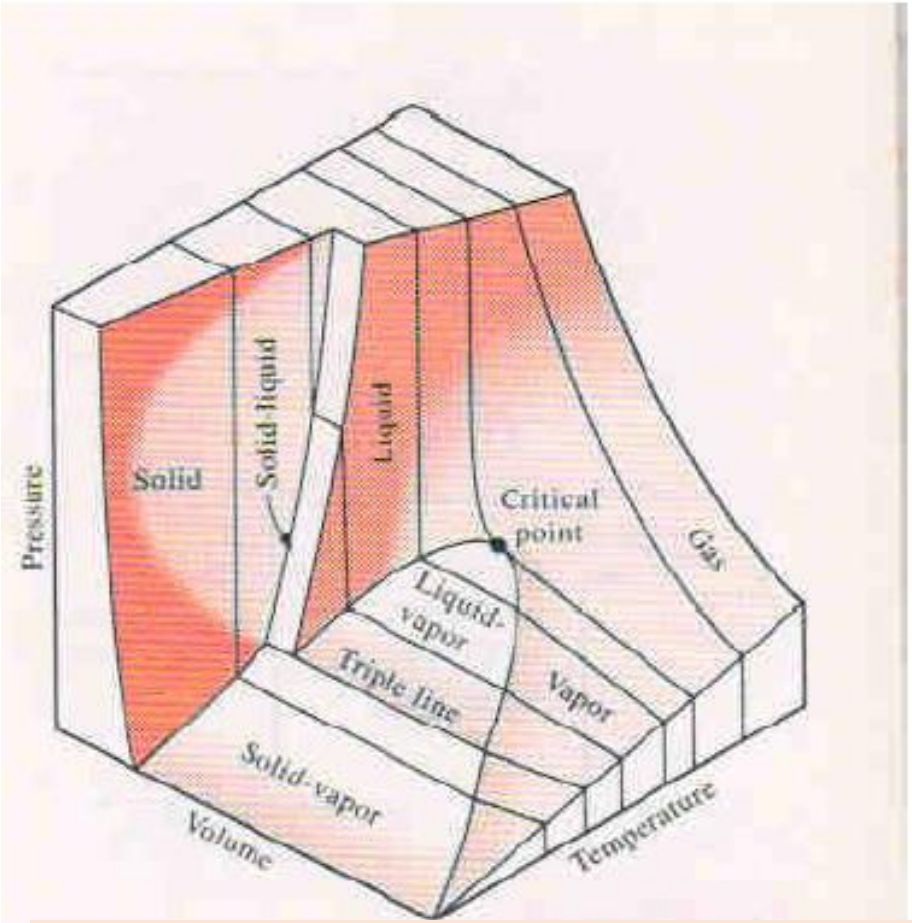




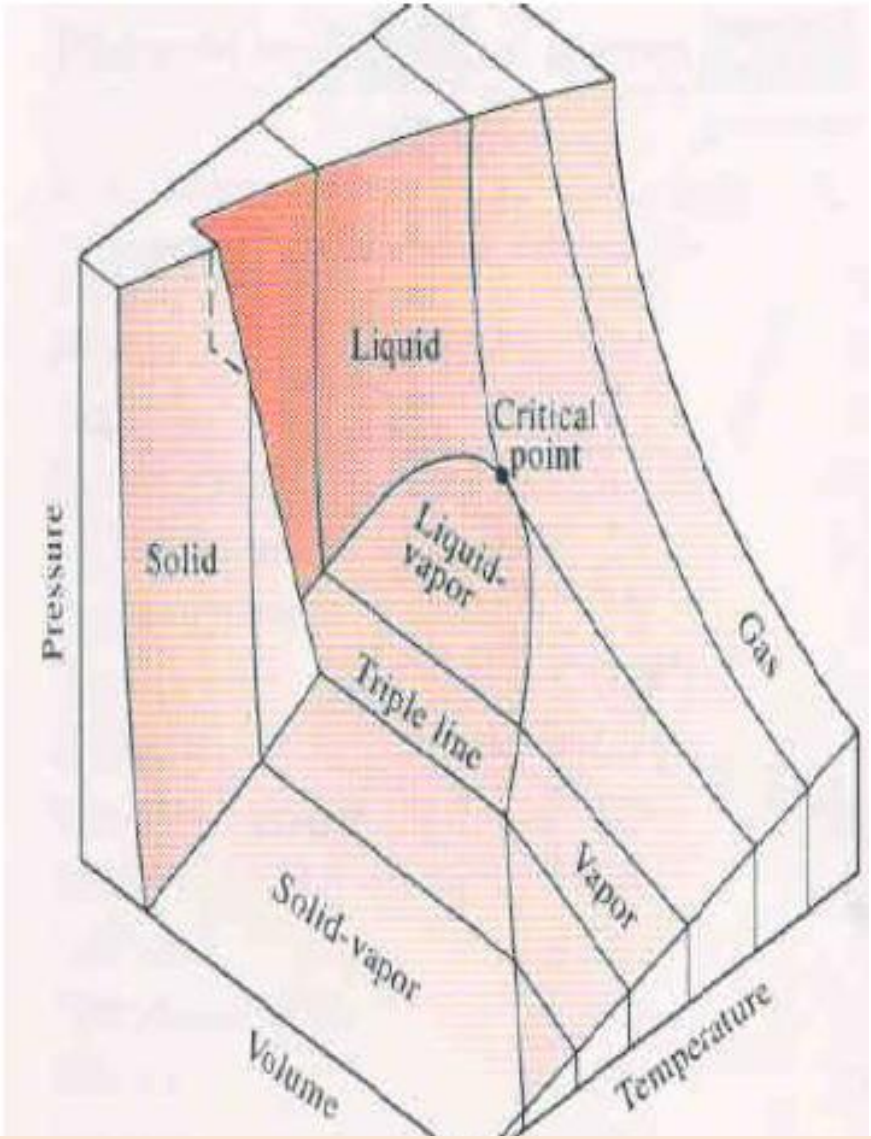
Change of Phase: depressurization at a constant temperature



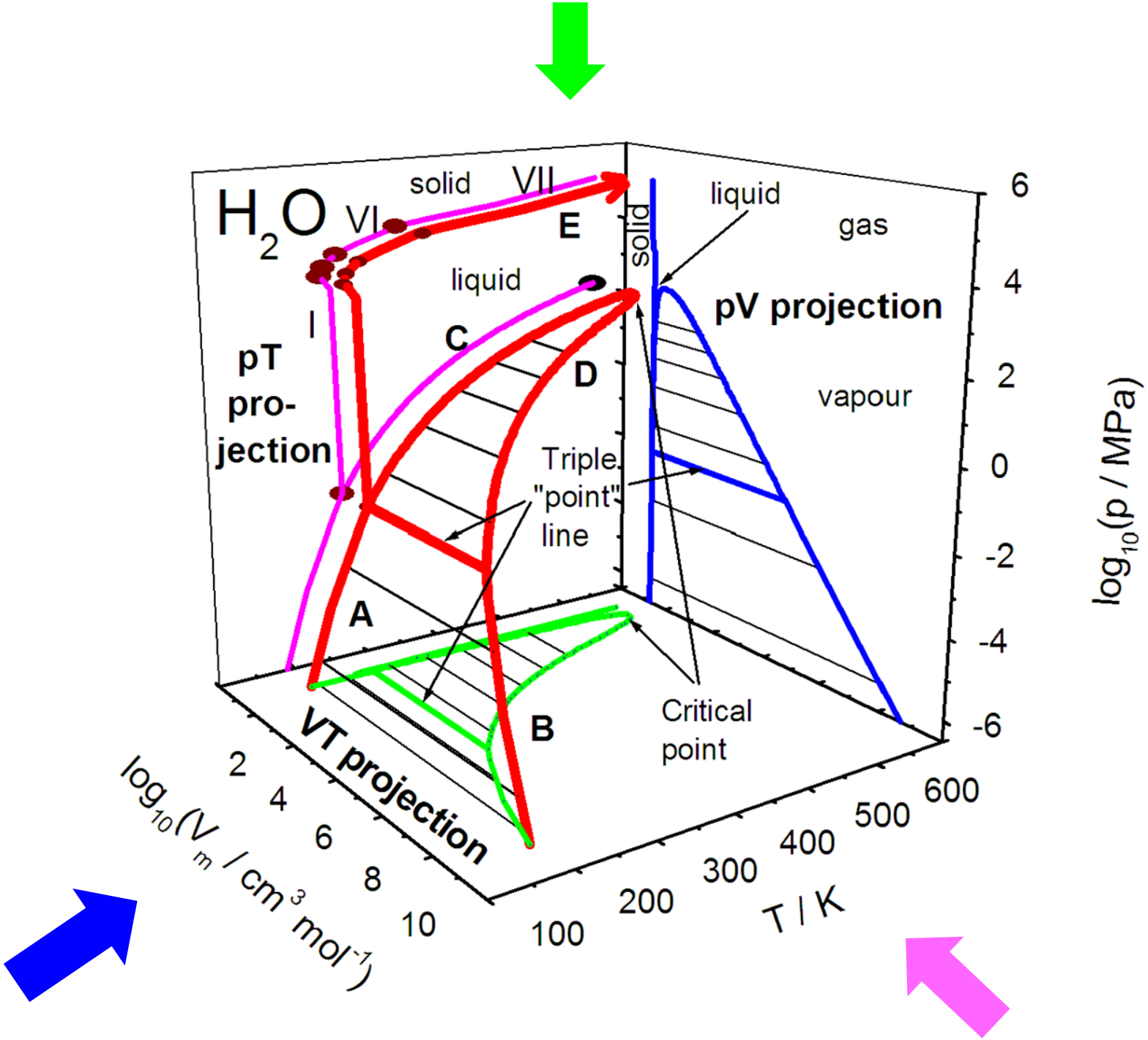




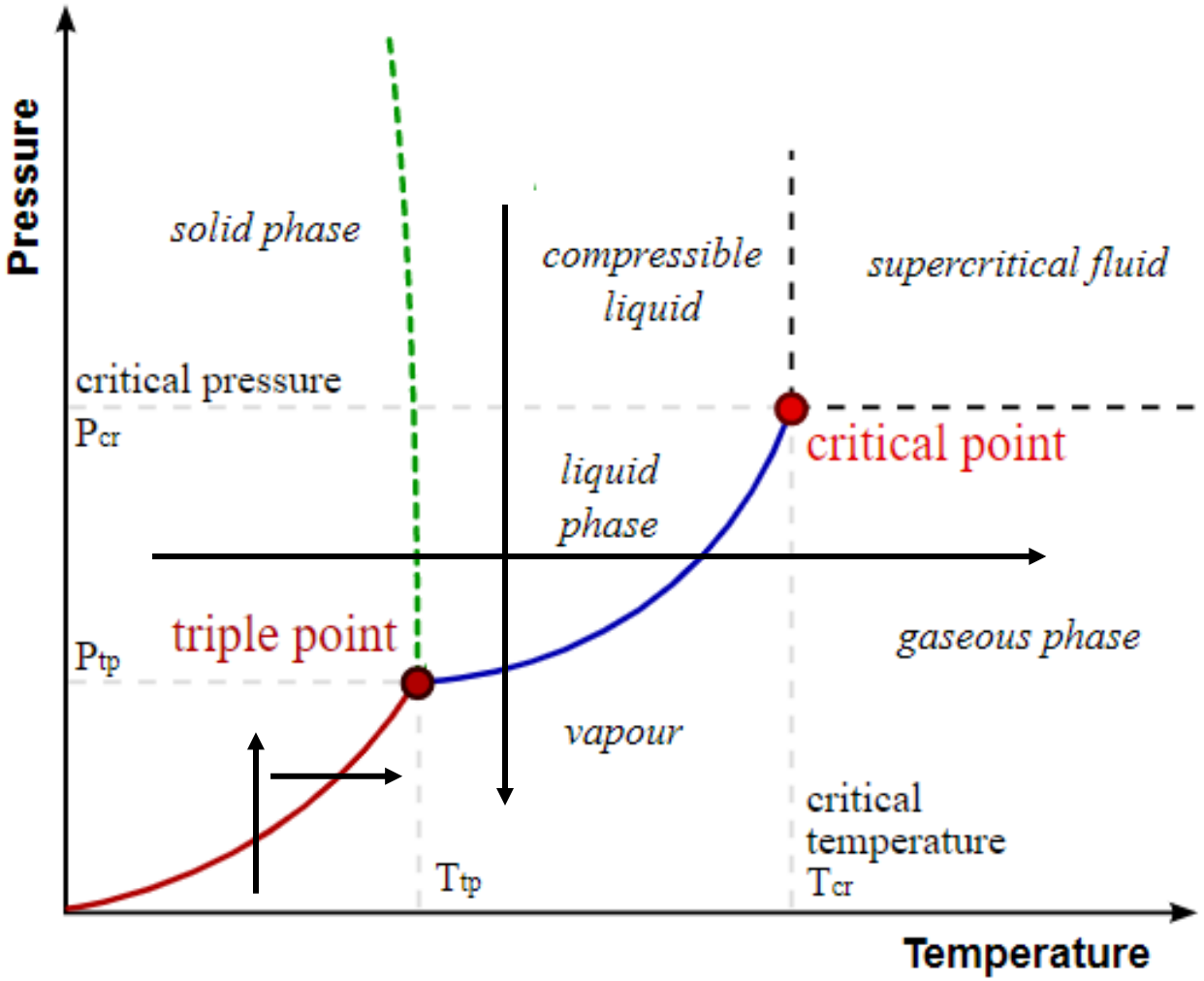
p-v-T surface of a substance that contracts on freezing



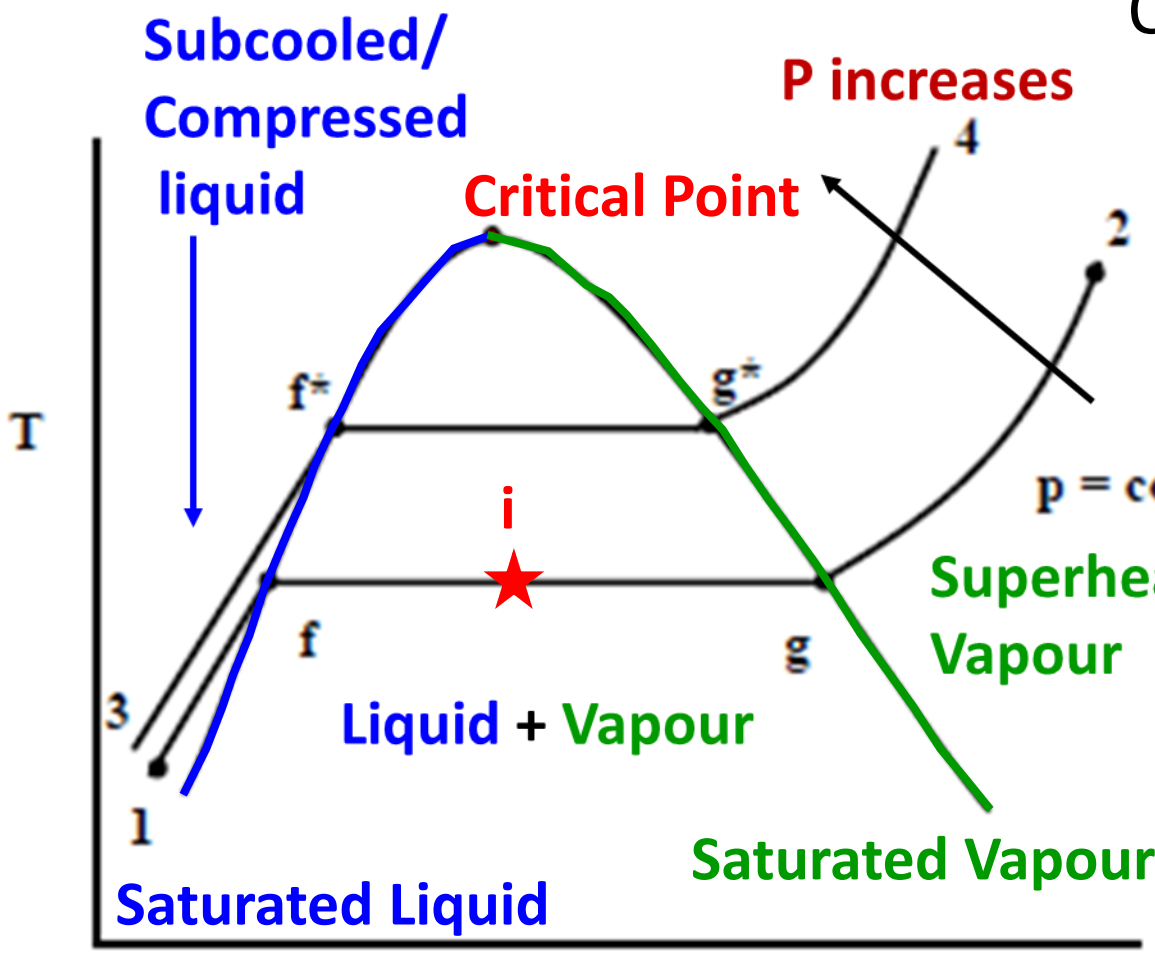
p-v-T surface of a substance that expands on freezing

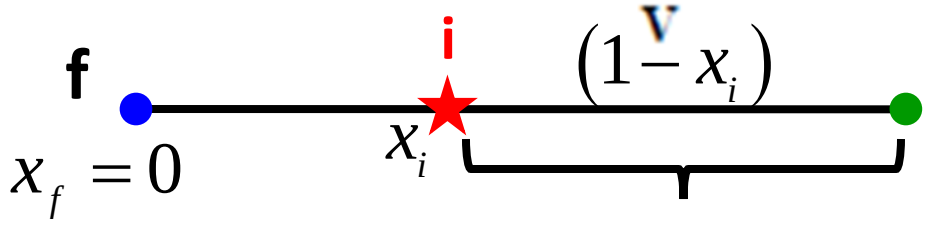

H2O
solid
VII
VI
E
liquid
I
C
D
pT
pro-
jection
A
B
VT projection
Triple
"point"
line
Critical
point
solid
liquid
gas
pV projection
vapour
log10(p / MPa)
6
4
2
0
-2
-4
-6
log10(Vm / cm3 mol-1)
2
4
6
8
10
T / K
100
200
300
400
500
600

Change of Phase: p-T curve



Change of Phase: Quality/Dryness fraction





Quality (dryness fraction)

$$x = \frac{m_g}{m_l + m_g}$$

$$x_f = 0;\ x_g = 1$$

$$v_i = x_i \times v_g + (1 - x_i) \times v_f$$

$$u_i = x_i \times u_g + (1 - x_i) \times u_f$$

$$h_i = x_i \times h_g + (1 - x_i) \times h_f$$

$$\frac{v_i - v_f}{v_g - v_f} = \frac{x_i - x_f}{x_g - x_f} = x_i$$

$$v_i = v_f + x_i\left(v_g - v_f\right)$$

$$= x_i v_g + (1 - x_i) v_f$$

Change of Phase: Use of Steam Table

Saturated Steam Temperature Table

T , °C	p, bar	v_f, m^3/kg	v_g, m^3/kg	u_f, kJ/kg	u_g, kJ/kg	h_f, kJ/kg	h_g, kJ/kg	s_f, kJ/kg.K	s_g, kJ/kg.K
0.01	0.0061	1.0002×10^{-3}	206.1	0.01	2376	0.01	2501	0.0	9.156
5.0	0.0087	1.0001×10^{-3}	147.1	21.00	2383	21.00	2511	0.0762	9.026
10.0	0.0123	1.0001×10^{-3}	106.4	42.01	2389	42.01	2520	0.151	8.901
15.0	0.0170	1.0007×10^{-3}	77.93	62.99	2396	62.99	2529	0.2245	8.781
20.0	0.0234	1.002×10^{-3}	57.79	83.94	2403	83.94	2538	0.2966	8.667
90.0	0.7013	1.036×10^{-3}	2.361	376.8	2494	376.9	2660	1.193	7.479
95.0	0.8455	1.039×10^{-3}	1.982	397.9	2501	398.0	2668	1.250	7.416
100.0	1.013	1.044×10^{-3}	1.673	418.9	2507	419.0	2676	1.307	7.355
110.0	1.433	1.052×10^{-3}	1.210	461.1	2518	461.3	2691	1.418	7.239
120.0	1.985	1.060×10^{-3}	0.892	503.5	2529	503.7	2706	1.528	7.130

$$v_i = x_i \times v_g + (1 - x_i) \times v_f$$

$$u_i = x_i \times u_g + (1 - x_i) \times u_f$$

$$h_i = x_i \times h_g + (1 - x_i) \times h_f$$

If x_i=0.3

$$v_i = 0.3 \times 1.673 + (1 - 0.3) \times 0.001044 = 0.5026\ m^3/kg$$

$$u_i = 0.3 \times 2507 + (1 - 0.3) \times 418.9 = 1045.33\ kJ/kg$$

$$h_i = 0.3 \times 2676 + (1 - 0.3) \times 419 = 1096.1\ kJ/kg.$$

Change of Phase: Use of Steam Table

Saturated Steam Pressure Table

p, bar	T, °C	v_f, m^3/kg	v_g, m^3/kg	u_f, kJ/kg	u_g, kJ/kg	h_f, kJ/kg	h_g, kJ/kg	s_f, kJ/kg.K	s_g, kJ/kg.K
0.02	17.50	1.001×10^{-3}	67.00	73.48	2400	73.48	2534	0.261	8.724
0.04	28.96	1.004×10^{-3}	34.80	121.4	2415	121.4	2554	0.423	8.475
0.06	36.15	1.006×10^{-3}	23.75	151.5	2425	151.5	2567	0.521	8.331
0.08	41.50	1.008×10^{-3}	18.1	173.8	2432	173.8	2577	0.593	8.229
0.1	45.8	1.010×10^{-3}	14.68	191.8	2438	191.8	2585	0.649	8.150
1.0	99.62	1.043×10^{-3}	1.694	417.3	2506	417.4	2675	1.303	7.359
1.5	111.4	1.053×10^{-3}	1.159	466.9	2520	467.1	2694	1.434	7.223
2.0	120.2	1.061×10^{-3}	0.886	504.5	2530	504.7	2707	1.530	7.127
3.0	133.6	1.073×10^{-3}	0.606	561.1	2544	561.5	2725	1.672	6.992
4.0	143.6	1.084×10^{-3}	0.463	604.3	2554	604.8	2739	1.777	6.896
5.0	151.9	1.093×10^{-3}	0.375	639.7	2561	640.2	2749	1.861	6.821

$$v_i = x_i \times v_g + (1 - x_i) \times v_f$$

$$u_i = x_i \times u_g + (1 - x_i) \times u_f$$

$$h_i = x_i \times h_g + (1 - x_i) \times h_f$$

If x_i=0.4

$$v_i = 0.4 \times 0.606 + (1 - 0.4) \times 0.001073 = 0.243\ m^3/kg$$

$$u_i = 0.4 \times 2544 + (1 - 0.4) \times 561.1 = 1354.3\ kJ/kg$$

$$h_i = 0.4 \times 2725 + (1 - 0.4) \times 561.5 = 1426.9\ kJ/kg.$$

Change of Phase: Use of Steam Table

Superheated Steam Tables

Pressure p = 1.0 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg.K
100.0	1.696	2507	2676	7.361
150.0	1.936	2583	2776	7.613
200.0	2.172	2658	2875	7.834
250.0	2.406	2733	2974	8.033
300.0	2.639	2810	3074	8.216
400.0	3.103	2968	3278	8.544
500.0	3.565	3132	3488	8.834
600.0	4.028	3302	3704	9.098
700.0	4.490	3479	3928	9.340
800.0	4.952	3664	4159	9.565
900.0	5.414	3855	4396	9.777

Pressure p = 2.0 bar				
T , °C	v, m^3/kg	u, kJ/kg	h, kJ/kg	s, kJ/kg.K
150.0	0.960	2577	2769	7.280
200.0	1.080	2654	2871	7.507
250.0	1.199	2731	2971	7.709
300.0	1.316	2809	3072	7.893
400.0	1.549	2967	3277	8.222
500.0	1.781	3131	3487	8.513
600.0	2.013	3301	3704	8.777
700.0	2.244	3479	3928	9.019
800.0	2.475	3663	4158	9.245
900.0	2.705	3855	4396	9.457

Use interpolation suitably to get properties like u, v, h as a function of (p, T)

Change of Phase: Use of Steam Table

If the condition pertains to sub-cooled liquid, then use temperature-based saturation table and write

$$h = u_f + pv_f$$ **At a temperature T_f**

This value of enthalpy will usually be very close to h_f

Thank You